



Note

On auction protocols for decentralized scheduling

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ABSTRACT

This note shows that, for the combinatorial market goods defined by Wellman et al. (2001), there may not exist an optimal allocation that is in equilibrium. Moreover, this result holds even if the value of each processed job is independent of its completion time. The proposed algorithm for finding an equilibrium allocation, using an assignment model, fails due to complementarity, even for instances that admit an optimal allocation which is in equilibrium.

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Wellman et al. (2001) define price equilibrium using the basic goods, time slots, as market goods. In equilibrium each agent, including the seller, receives an allocation that maximizes its utility given the current prices. Under this definition, it is shown that (i) for general discrete resource allocation problems, if there exist prices that support an equilibrium solution, then this solution is optimal; and (ii) equilibrium prices may not exist.

The problem we consider is defined by adopting the notation used by Wellman et al. (2001), with minor modifications. We define

- G , a set of n discrete *basic goods*; a basic good is also referred to as a *time slot*;
- A , a set of m agents, and $\perp$ representing the seller or null agent;
- q_i , the value received by the seller for holding time slot i , i.e. the reserve value of time slot i ;
- f , a *solution* $G \rightarrow A \cup \{\perp\}$ indicating which agent, if any, receives each time slot;
- $F_\perp$, the set of unallocated time slots in f ;
- $F_j \equiv \{i \mid f(i) = j\}$, a set of time slots allocated to agent j in f ;
- G' , a set of *market goods*, where a market good is a pair (y, z) , denoting a bundle of y time slots containing time slot z and an indeterminate set of $y - 1$ slots strictly before z ;
- $p(y, z)$, the price of market good (y, z) ;
- ϕ , a *market allocation* $G' \rightarrow A \cup \{\perp\}$ assigning market goods to agents;
- $\Phi_\perp$, the set of unallocated market goods in ϕ ;
- $\Phi_j \equiv \{(y, z) \mid \phi((y, z)) = j\}$, the set of market goods allocated to agent j in ϕ .

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A market allocation ϕ and a solution f are *consistent* if f provides each agent what is promised by ϕ . That is, for all $j \in A$ and $k \leq n$, we have

$$|G_k \cap F_j| = \sum_{i \leq k} \begin{cases} y, & \text{if } (y, i) \in \Phi_j, \\ 0, & \text{otherwise,} \end{cases}$$

where $G_k \equiv \{1, \dots, k\}$ is the set of time slots with index no larger than k . Observe that, given n basic goods, the number of possible market goods (y, z) is $O(n^2)$.

Each agent j possesses a job with processing time λ_j , i.e., a minimum of λ_j time slots are required to process the job of agent j . However, preemption of job processing is allowed, and hence the required λ_j time slots are not necessarily consecutive. The value received by agent j is a nonincreasing function of its job completion time, where we use v_j^c to denote this value if agent j 's job is processed with completion time c . An agent receives 0 value if its job is not processed. We further define

- $v_j(X)$, the value received by agent j for holding the set of time slots $X \subseteq G$, which is equal to 0 if $|X| < \lambda_j$, and is otherwise equal to v_j^k where k is the earliest time slot such that X contains λ_j time slots no later than k ;
- $v(f) \equiv \sum_{i \in F(\perp)} q_i + \sum_{j \in A} v_j(f_j)$, the *global value* of the solution f ;
- $v'_j(y, z) = \begin{cases} v_j^z, & \text{if } y \geq \lambda_j, \\ 0, & \text{otherwise,} \end{cases}$ the value agent j can achieve by using market good (y, z) .

A solution is *optimal* if it maximizes the global value, and a market allocation is *optimal* if it is consistent with an optimal solution.

Let $Y(\Phi, c)$ denote the number of time slots no later than c in a set Φ of market goods. The maximum surplus that agent j can ensure by purchasing market goods at prices p is given by

$$H'_j(p) = \max_{\Phi} \left\{ \max_c \{v'_j(Y(\Phi, c), c)\} - \sum_{(y, z) \in \Phi} p(y, z) \right\}.$$

A market allocation ϕ is defined by Wellman et al. (2001) to be in *equilibrium* at prices p iff

1. for each agent j , $\max_c \{v'_j(Y(\Phi_j, c), c)\} - \sum_{(y, z) \in \Phi_j} p(y, z) = H'_j(p)$;
2. for all (y, z) , $p(y, z) \geq \min_{B \subseteq G_z: |B|=y} \{\sum_{i \in B} q_i\}$;
3. there exists a solution f , consistent with ϕ , such that
 - (i) for all j , $\sum_{(y, z) \in \Phi_j} p(y, z) \geq \sum_{i \in F_j} q_i$;
 - (ii) for all (y, z) , if $|G_z \cap F_{\perp}| \geq y$, then

$$p(y, z) \leq \min_{\{B \subseteq G_z \cap F_{\perp}: |B|=y\}} \left\{ \sum_{i \in B} q_i \right\}.^1 \quad (1)$$

The first condition requires that each agent maximizes its surplus at the given prices. The second condition ensures that prices are at least high enough that the seller would in principle be able to construct a market good from basic goods. The third condition ensures that (i) the allocated market goods are priced no lower than the reserve price, and that (ii) the market goods that could be constructed from the unallocated basic goods are priced no higher than the minimum reserve prices.

This definition of price equilibrium is unconventional. It does not require that the allocation maximizes the seller's revenue. Instead, it only ensures that the seller would be willing to part with each market good individually. If the former condition were enforced, then optimality would be guaranteed by standard arguments such as in Bikhchandani and Ostroy (2002). However, existence would still be in question because the valuations may exhibit both complements and substitutes. For example, a bundle of time slots can be either more or less valuable to an agent, depending upon which other bundles of time slots that agent possesses.

Wellman et al. (2001, Theorem 7.2) state that, for a given set of market goods G' , if $q_i \geq q_{i'}$ for all $i \leq i'$, then there exists an equilibrium allocation ϕ that is consistent with a solution which maximizes the global value. The condition $q_i \geq q_{i'}$ for all $i \leq i'$ defines monotone reserve prices, which is consistent with the feature of the problem studied that an earlier time slot is more valuable to agents than a later time slot. However, the following example contradicts this theorem, even for the case where the value of each processed job is independent of its completion time.

¹ In Wellman et al. (2001), Eq. (1) contains a " $\geq$ " instead of a " $\leq$ " sign. We view this as a typo, based on our understanding of the problem, and a subsequent statement in Wellman et al. (2001) that "when a market good could be satisfied by basic goods unallocated in this solution, the reserves of those goods define an upper bound on its price".

Example 1. The seller owns two time slots, 1 and 2, with reserve values 5 and 4, respectively. There are two job agents, with $\lambda_1 = 1$, $\lambda_2 = 2$, $v_1 = 12$ and $v_2 = 15$. Here the value of a processed job is independent of its completion time, so we omit the superscripts on v_1 and v_2 .

As can be verified, the unique optimal solution, denoted by f^* , has $F_\perp^* = \{1\}$, $F_1^* = \{2\}$ and $F_2^* = \emptyset$, and achieves a global value of $5 + 12 + 0 = 17$.

There is a unique market allocation ϕ that is consistent with solution f^* , where $\Phi_\perp^* = \{(1, 1), (2, 2)\}$, $\Phi_1^* = \{(1, 2)\}$, and $\Phi_2^* = \emptyset$. Moreover, f^* is the unique solution that is consistent with ϕ^* . Therefore, to prove the nonexistence of an optimal allocation that is in equilibrium for the given instance, it suffices to show that ϕ^* is not in equilibrium.

If ϕ^* is in equilibrium, then from Eq. (1), we have $p(1, 1) = 5$. We consider two cases for $p(1, 2)$. First, if $p(1, 2) > 5$, then agent 1 can gain more value by buying (1, 1) instead of (1, 2). Second, if $p(1, 2) \leq 5$, then the total price of (1, 1) and (1, 2) is no greater than 10, and agent 2 can gain more value by purchasing both market goods. In either case, the market allocation is not in equilibrium.

Wellman (2007) acknowledges that the proof of Theorem 7.2 is incorrect. In the solution approach, a competitive equilibrium for a related assignment problem is first constructed, and is then adjusted to provide an equilibrium allocation for the scheduling problem. In the proof, it is stated that “Since these elements are priced in an assignment equilibrium, we know that the agents would not prefer to choose an alternative.” However, an assignment equilibrium does not rule out the possibility that an agent may wish to switch to multiple other goods if they are complements. Thus, to guarantee the existence of an assignment equilibrium, complementarity must not exist. However, in the scheduling problem studied, complementarity may exist.

In fact, the solution approach may fail to find an equilibrium allocation, even for instances that admit an optimal allocation which is in equilibrium. The following example illustrates this observation.

Example 2. The seller has two time slots to sell, both with reserve value 0. There are three agents, each with a job, where $\lambda_1 = \lambda_2 = 1$, $\lambda_3 = 2$, $v_1 = v_2 = 2$ and $v_3 = 3$. As in Example 1, the value of a processed job is independent of its completion time.

The optimal solution is to allocate one time slot to agent 1 and the other time slot to agent 2, thus achieving a global value of 4. As can be verified, the market allocation ϕ with $\Phi_\perp = \{(2, 2)\}$, $\Phi_1 = \{(1, 1)\}$, $\Phi_2 = \{(1, 2)\}$, $\Phi_3 = \emptyset$, $p(1, 1) = 1.5$, $p(1, 2) = 1.5$, and $p(2, 2) = 3$ is consistent with this solution.

In the associated assignment problem constructed, the set of source entities contains the three agents and the seller, and the target set contains market goods (1, 1) and (1, 2), a null market good, and the seller. The price vector (0, 0, 0, 0) is the minimum equilibrium price that supports an optimal assignment. Thus, prices 0, 0, and 0 are obtained for market goods (1, 1), (1, 2) and (2, 2). However, these prices do not support an equilibrium allocation, since agent 3 can gain more value by offering a higher price to buy both market goods (1, 1) and (1, 2).

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